

Introduction

Since the turn of the millennium, the number of Internet users has exploded around the world. The constant global expansion of the Internet backbone, the never-ending improvement in Internet speed, and the ongoing price reductions in Internet service have resulted in the Internet being utilized in practically every household around the globe. Not to add the rapid advancement of mobile Internet and the gadgets those support it, such as smartphones, which has resulted in Internet access being available practically anywhere on the planet. In the meantime, online video sharing took place. As a result, several online video-sharing platforms, such as YouTube, arose and quickly acquired popularity. However, due to the COVID-19 pandemic that broke out, a complete school system was compelled to shift from classrooms to the Internet.

The coronavirus pandemic had resulted in an unparalleled surge. The spread of COVID-19 posed a threat to humanity, as the pandemic had compelled numerous worldwide activities, including educational institutions, to shut down. Despite the challenges posed by this rapid transformation, educational institutions had been pushed to move to e-learning using accessible educational platforms in order to prevent the virus from spreading. Because of the unexpected closure of educational institutions as a result of COVID-19's emergence, authorities have proposed that in an emergency, alternate learning methods be used to ensure that students are not left without a way to study and the pandemic does not spread. While education can take place within (or outside) the classroom, the use of computer technology and the Internet is a crucial component of e-learning. E-learning is an important part of today's educational system because it alters the overall structure and is now one of the most popular themes amongst academics. An online educator compensates for the virtual classroom's absence of physical presence by providing a welcoming environment in which everyone feels comfortable participating in class. It contributes significantly to the transformation of our life into the digital world by providing flexible locations, class times, and high-quality content. It presents a path for humanity's progress during a period of physical isolation caused by a pandemic.

We may adapt our procedures and processes based on the demands of the learners using e-learning technologies. There are numerous online tools accessible, all of which are essential for a productive and efficient learning environment. In this time of crisis, educators can reach out to their students using a combination of audio, video, and text to keep their lectures personable. This can aid in the creation of a collaborative and dynamic learning environment in which students can provide rapid feedback, ask questions, and learn in an engaging manner. In times of crisis, such as man-made calamities, natural disasters, or pandemics like Covid-19, the Anywhere-Anytime characteristic of e-learning comes in handy. Closures of places and risky road trips can cause a lot of problems, but e-learning will allow us to acquire an education from the comfort of our own homes or workplaces. Technology allows individuals to connect and even work remotely without the need for face-to-face interaction during times of crisis, providing inventive and resilient solutions to prevent disruption. E-learning has flaws in the sense that it might obstruct contact between the learner and the educator, resulting in a loss of direct communication and personal touch. Many technical issues might obstruct and slow down the teaching-learning process for users. Students and learners are not all the same; their abilities and levels of confidence differ. Some people are uncomfortable learning online, which leads to irritation and uncertainty. Inadequate compatibility

between the design of technology and the psychological components required by the learning process, as well as insufficient customisation of learning processes, can block the teaching process and produce an imbalance.

Covid pandemic crisis ushered in a new era for online learning, allowing individuals to consider the positive aspects of e-learning technologies. This was a period when there was a lot of room for new discoveries and digital developments to emerge. Users of any age can utilize the online resources in critical scenario like Covid pandemic and benefit from the time and place flexibility that comes with online learning.

While digital or online education cannot completely replace traditional classroom instruction, it does have significant advantages. It enables for flexible and personalised learning at the learner's own pace, as well as the ability to continually enhance and expand knowledge via digital means. The rapid growth of internet usage, as well as many government programmes such as the Digital India campaign, have produced an atmosphere that is favourable to the transition to digital education. This will be supplemented by the Ministry of Human Resource Development's (MHRD) recent announcement of PM e-Vidya, a nationwide programme that will unite all activities connected to digital/online/on-air education. DIKSHA (one nation – one digital platform), TV (one class – one channel), SWAYAM (online MOOCS on various topics), IITPAL (platform for exam preparation), AIR (through community radio and the CBSE Shiksha Vani podcast), and AIR (through community radio and the CBSE Shiksha Vani podcast) and NIOS-developed study materials for differently abled children are among these. MHRD has also come up with an innovative idea which has been named as 'PRAGYATA' (guidelines for digital and online education) consisting of mainly eight steps: Plan, review, arrange, guide, yak (talk), assign, track and appreciate. The MHRD will extend and develop all of these areas of e-learning in a gradual manner, in a methodical and consistent manner.

Need and Rationale of the Study

The COVID-19 has resulted in schools shut all across the world. Globally, over 1.2 billion children are out of the classroom. As a result, education has changed dramatically, with the distinctive rise of e-learning, whereby teaching is undertaken remotely and on digital platforms. Although there are many prospects for online learning in general, but pandemic time played a great role to flourish it, as most academic institutions have converted to this format. During the Corona Virus outbreak, online learning, remote working, and e-collaborations skyrocketed. Our Delhi Government has taken many initiatives to provide online education via different LMS platforms to access the students to continue their learning. Various steps have also been taken to provide educational material to those who do not have the facility of online mode. This research has been taken to find the perspective of different stakeholders mainly: HoS, teachers, students and parents regarding online education with respect to various aspects such as their perception regarding virtual teaching and learning, advantages and disadvantages of virtual mode, issues and challenges faced by different stakeholders and so on.

Statement of the Problem

Keeping the above-mentioned facts in mind, the investigator decided to conduct a research on

“A study of the perception of different stakeholders towards virtual teaching-learning mode during Covid-19 pandemic at upper primary stage”

Operational Definition of the Key Terms Used in The Study

Stakeholders: a sort of organization or system in which all members or participants are considered as having the same interest and role in its success. In the education system, stakeholders include administrators, teachers, staff members, students, parents, families, community members, local business leaders, and elected officials such as school board members, and state representatives who are invested in the welfare and success of a school and its students.

Virtual education: an electronic aided learning that uses the internet and technological tools for the acquisition and transmission of knowledge.

Pandemic: a widespread occurrence of an infectious disease over a large geographic area (such as many countries or continents), often impacting a significant proportion of the population.

Covid-19: a coronavirus-caused acute sickness in humans that is marked by fever and cough and can escalate to severe symptoms and, in some cases, death, especially in the elderly and those with underlying health issues.

Research Questions

The study was guided with the following research questions:

1. What are the different teaching methodologies used by the teacher to deliver content online at the upper primary stage?
2. What are the issues and challenges faced by different stakeholders in the teaching learning mode during covid-19 pandemic?
3. What are the experiences of students concerning online learning during Covid-19?
4. What is the perception of different stakeholders with respect to advantages and disadvantages of virtual teaching learning mode?

Objectives of the Study

The proposed study will serve the following objectives:

1. To explore various forms of virtual teaching-learning modes used by the teachers of upper primary stage during covid-19 pandemic.
2. To study the perception of different stakeholders with respect to virtual teaching and learning.
3. To explore various challenges faced by different stakeholders in adapting to the virtual teaching-learning process during covid-19 pandemic.

Delimitation of the Study

The present study was confined to:

- ❖ Schools of Directorate of Education (DoE), GNCT of Delhi
- ❖ District East
- ❖ Zone: 1, 2 and 3

Conceptual Framework

Because of Covid-19, countries all over the world had been put on lockdown, and educational institutions had been no exception. The country was experiencing huge crises in several areas as a result of the Covid-19 outbreak across the country, but the education sector was the worst hit. Educational institutions should use innovative approaches to combat the chronic crisis like covid-19 pandemic. During the lockdown period, many Indian state governments were looking into using an online learning platform to access online educational resources. They were trying to come up with strategies to help students keep up with their studies while the country was shut down because to Covid-19. The goal was to keep pupils' academic cycles on track and not let the lockout interfere with their study. The MHRD encourages students to continue their studies using online learning systems to ensure that there is no interruption in their education and that they have full access to classes as before. Students can use online learning platforms to not only have complete access to study materials, but also to participate in online classes and interact with teachers in the same way they would in a real classroom setting. Here are some online platforms initiated by MHRD to provide free access to online educational resources.

SWAYAM online courses: provides access to the top educational learning resources that were previously only available on the SWAYAM Platform and are now available to any learner for free and without the need to register. There are 1900 entire courses available, including teaching videos, weekly assignments, tests, and credit transfers, geared for students in grades 9 through 12, as well as those in higher education (undergraduate and postgraduate). Engineering, humanities, social sciences, law, and management courses, including robotics, are all connected with the curriculum.

SWAYAMPRAKASHA: is a group of 32 DTH channels that provide high-quality educational curriculum-based course contents in a variety of disciplines, including arts, science, commerce, performing arts, social sciences and humanities subjects, engineering, technology, law, medicine, agriculture, and other subjects to all teachers, students, and citizens across the country who are interested in lifelong learning. Courses for school education (grades 9-12), higher education (undergraduate and postgraduate), out-of-school children, vocational education, and teacher training are broadcast on the channels. Arts, science, commerce, performing arts, social sciences, humanities, engineering, technology, law, medicine, and agriculture are among the subjects covered. On the website, one can see television broadcast schedules as well as archival shows.

National Digital Library: All academic levels, including researchers and life-long learners, all disciplines, all popular forms of access devices, and differently-abled learners can access the National Digital Library, which is a digital repository of a vast amount of academic content in various formats and provides interface support for leading Indian languages for all academic levels, including researchers and life-long learners, all disciplines, all popular forms of access devices, and differently-abled learners.

DIKSHA: The DIKSHA portal contains curriculum-aligned online learning content for students, instructors, and parents, including as video courses, worksheets, textbooks, and assessments. The content was generated by more than 250 instructors who teach in different languages under the supervision of the country's national board of education (CBSE) and the National Council of Educational Research and

Training (NCERT). Students are encouraged to venture beyond the book through QR codes in textbooks. Offline use is possible with the app.

e-Pathshala: NCERT's e-Pathshala is a multilingual online learning tool for students in grades 1 to 12. In many languages, including Hindi, Urdu, and English, the app contains books, videos, audio, and other content intended towards students, educators, and parents.

The National Repository of Open Educational Resources (NROER): This portal offers a variety of resources in many languages for students and teachers, including books, interactive modules, and videos, as well as a variety of STEM-based games. For classes 1–12, content is mapped to the curriculum, and teachers' materials are aligned.

Dealing with COVID-19, the education system's shared vision acknowledged that throughout the pandemic, instructors and students are motivated to modify online teaching-learning platforms to meet contemporary educational needs. Everyone, whether professors or students, was friendly and adept in using social media apps such as WhatsApp, Facebook, Twitter, and Instagram, which facilitated the use of online educational platforms such as ZOOM, Cisco WebEx, Google Meet, and others as an indication of positive learning transfer. There are also several valuable educational apps available, including Office 365, Google Classroom, and a variety of other user-friendly video conferencing software, all of which can be downloaded for free and are simple to use.